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DIFFERENCE BETWEEN STACK AND HEAP MEMORY IN JAVA

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- JVM has divided memory space between two parts: one is Stack and another one is Heap space. Stack space is mainly used for storing order of method execution and local variables.
- Stacks always store blocks in LIFO order whereas heap memory uses dynamic allocation for allocating and deallocating memory blocks.
- Memory allocated to the heap lives until one of the following events occurs :
 - Program terminated
 - Memory free

WHAT IS A HEAP MEMORY?

- Heap memory is allocated for storing objects, arrays, and JRE (Java Runtime Environment) classes. Memory may be allocated at random address and automatically free memory that is not in use.



THE FOLLOWING IS AN EXAMPLE OF A HEAP IN JAVA:

```
public class HeapExample {  
    public static void main(String[] args) {  
        int[] numbers = new int[3]; // Array object in heap  
        numbers[0] = 5;  
        numbers[1] = 10;  
        numbers[2] = 15;  
        for (int num : numbers) {  
            System.out.println(num);  
        }  
    }  
}
```

The output of the above Java program is:

**5
10
15**

WHAT IS A STACK MEMORY?

- Stack memory is used to store items that have a very short life, such as local variables and reference variables of objects. It follows the LIFO (Last-In First-Out) principle and is automatically managed during method calls and exits.

```
public class StackExample {  
    public static void main(String[] args)  
    {  
        int a = 10;    // stored in stack  
        int b = 20;    // stored in stack  
        int sum = add(a, b); // method  
call adds a new frame on the stack  
        System.out.println("Sum: " + sum);  
    }  
  
    public static int add(int x, int y) {  
        int result = x + y; // local  
variable in stack  
        return result;  
    }  
}
```

Sr. No.	Key	Stack	Heap Memory
1	Basic	Stack memory is used to store items which have a very short life like local variables, a reference variable of objects	Heap memory is allocated to store objects and JRE classes.
2	Ordering	The stack is always reserved in a LIFO (last in first out) order	Heap memory is dynamic allocation there is no fixed pattern for allocating and deallocating blocks in memory
3	Size	We can increase stack memory size by using JVM parameter -XSS	We can increase or decrease heap memory size by using JVM option -Xms and -Xmx
4	Visibility	Variables are visible to only to owner thread	It is visible to all threads
5	Exception	JVM will throw <code>java.lang.StackOverFlowError</code>	JVM will throw <code>java.lang.OutOfMemoryError</code>

SUMMARY

- In Java, **stack memory** is used for short-lived data like method calls, local variables, and object references, operating on a Last-In, First-Out (LIFO) basis. **Heap memory** is where all objects and instance variables are stored, is shared across the entire application, and is managed by the Java Garbage Collector.